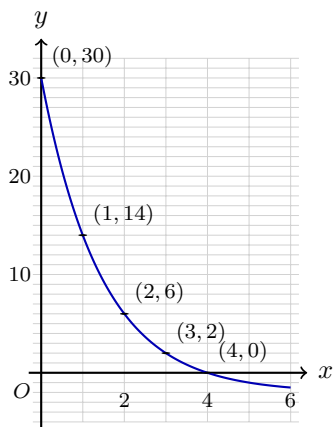


6 Topic Assessment Form B

1. What is the solution to the equation $\log_3(2x-3) = -1$? Round to the nearest thousandth.

$x =$

2. The graph shows the function $f(x) = 32\left(\frac{1}{2}\right)^x - 2$. What is the value of the inverse function f^{-1} at $x = 2$?



$f^{-1}(2) =$

3. What function is the inverse of the exponential function $y = \left(\frac{1}{3}\right)^x$?

- Ⓐ $y = x^{\frac{1}{3}}$
 Ⓑ $y = \left(\frac{1}{3}\right)^x$
 Ⓒ $y = \log_{\frac{1}{3}} x$
 Ⓓ $y = \log_x \left(\frac{1}{3}\right)$

4. What is the asymptote of the graph of $g(x) = \log_2(x+4)$?

- Ⓐ vertical at $x = 0$
 Ⓑ vertical at $x = -4$

- Ⓒ horizontal at $y = 0$
 Ⓓ horizontal at $y = -4$

5. Complete the equation of the inverse of the function $f(x) = \log_4(3x)$.

4 $\frac{1}{3}$ $\frac{1}{4}$ 3

$f^{-1}(x) =$ $\left(\right.$ $\left.)^x$

6. Use the properties of logarithms to choose the expression $x \log 2 - \log \frac{1}{3}$ as a single logarithm.

- Ⓐ $\log\left(2x - \frac{1}{3}\right)$
 Ⓑ $\log\left(\frac{2}{3x}\right)$
 Ⓒ $\log\left(2^x - \frac{1}{3}\right)$
 Ⓓ $\log(3(2)^x)$

7. The profit for a new website, in thousands of dollars, can be represented by 5^x , where x is time in months. Find the time when the company's profit is \$15,000. Select the exact solution as a logarithm, and select an approximate solution.

- ☐ Ⓐ. $x = \log_{15} 5$
☐ Ⓑ. $x = \log_5 15$
☐ Ⓒ. $x = \log 15$
☐ Ⓓ. $x = 1.689$
☐ Ⓔ. $x = 0.594$
☐ Ⓕ. $x = 0.176$

8. Complete the steps to write the expression $2\log a + 2\log b - 3\log c$ as a single logarithm using the properties of logarithms.

Product

Quotient

Power

$$2\log a + 2\log b - 3\log c$$

$$= \log a^2 + \log b^2 - \log c^3$$

$$= \log a^2 b^2 - \log c^3$$

$$= \log \left(\frac{a^2 b^2}{c^3} \right)$$

Addendum — Teacher Answer Key

Topic 6 Assessment, Form B • enVision® Algebra 2

1. $x = \boxed{1.667}$
 $\log_3(2x - 3) = -1 \Rightarrow 2x - 3 = 3^{-1} = \frac{1}{3} \Rightarrow 2x = \frac{10}{3} \Rightarrow x = \frac{5}{3} \approx 1.667.$
2. $f^{-1}(2) = \boxed{3}$
From the graph, $f(3) = 2$; therefore $f^{-1}(2) = 3$.
3. **C** $y = \log_{\frac{1}{3}} x$
The inverse of $y = b^x$ is $y = \log_b x$.
4. **B** vertical at $x = -4$
A log function $\log_2(x + 4)$ is undefined when $x + 4 = 0$, giving a vertical asymptote at $x = -4$.
5. $f^{-1}(x) = \boxed{\frac{1}{3}}(\boxed{4})^x$
If $y = \log_4(3x)$, then $4^y = 3x$, so $x = \frac{1}{3}(4)^y$. Swapping variables: $f^{-1}(x) = \frac{1}{3}(4)^x$.
6. **D** $\log(3(2)^x)$
 $x \log 2 - \log \frac{1}{3} = \log 2^x - \log \frac{1}{3} = \log\left(\frac{2^x}{1/3}\right) = \log(3 \cdot 2^x).$
7. **B** $x = \log_5 15$ and **D** $x \approx 1.689$
 $5^x = 15 \Rightarrow x = \log_5 15$, which is approximately 1.689 as rounded in the answer key.
8. Steps (in order): **Power, Product, Quotient.**

$$\begin{aligned} 2 \log a + 2 \log b - 3 \log c &= \log a^2 + \log b^2 - \log c^3 && \text{(Power)} \\ &= \log(a^2 b^2) - \log c^3 && \text{(Product)} \\ &= \log\left(\frac{a^2 b^2}{c^3}\right) && \text{(Quotient)} \end{aligned}$$

Answer key reflects Savvas enVision Algebra 2, Topic 6 Assessment Form B, Assessment Sourcebook. Explanations are supplemental notes for instructor reference.